Kshetrajna

Private Adaptive Intelligence for Windows
Team SankatMochan / Hackathon prototype v0.3

Project overview

Kshetrajna is a local Windows workload and file assistant that connects app activity, system pressure and relevant files in one dashboard. It learns recurring app routines, explains recommendations and keeps the user in control of changes. The name draws on the Sanskrit idea of the "knower of the field."

Problem and proposed value

Users inspect resource usage, switch between applications and search for related files through separate tools. Kshetrajna uses local context to help them identify which apps matter now and which files may support the current workflow. Its intended benefit is less manual investigation. Controlled experiments to measure that benefit remain future work.

Implemented prototype

The application collects real CPU/RAM measurements, process working sets, foreground-app switches and idle activity after opt-in. Session-based sequence learning finds recurring app groups and ordered routines. Next-app predictions use recent transition history and abstain when evidence is weak. The dashboard explains resource suggestions and supports saved app boards. Optional, individually approved CPU-priority trials last 60 seconds, with a durable journal, undo and restart recovery.

File intelligence indexes user-approved folders with metadata and bounded text excerpts. It provides keyword search, basic category/tag suggestions and contextual file recommendations. A reviewed rename/move backend/API records proposals and outcomes and supports conflict-aware undo. The organizer UI is a next step.

Privacy and architecture

Python 3.13 and its standard library call Win32 through ctypes. SQLite stores telemetry and a separate file catalog locally. HTML/CSS/JavaScript provide the browser dashboard through a loopback HTTP server. There are no third-party runtime packages or required cloud services. Collection defaults to off and includes pause, retention and deletion controls. Telemetry excludes window titles, URLs and keystrokes. Approved folder indexing can store short text excerpts. The current SQLite databases are not application-encrypted.

Validation and Snapdragon device

All **61 automated tests passed on the Snapdragon ARM64 laptop** in the project owner's device run. The suite also passed on Windows x64. Tests cover privacy, storage, workflow prediction, action recovery, file operations and HTTP integration. The device is an HP laptop with **Snapdragon X (processor label X126100), 16 GB RAM and an ARM-based 64-bit system**. The prototype does not use the NPU. Controlled performance measurements remain future work.

Run and repository

Run **run_demo.cmd** for clearly labeled synthetic scenarios that simulate approvals. Run **run.cmd** for live Windows observation, then enable collection. Next steps are device diagnostics, the organizer panel and controlled performance trials.
https://github.com/Team-SankatMochan/KSHETRAJNA-OS